Supplementary Material: How Robust Are Foundation-Model Camouflage Detectors?

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I. Statistical tests

Table S1 lists the paired bootstrap tests behind the comparisons in the main paper. For each image we average S_α over the conditions of a group, take the difference between two models on the same images, and resample these differences 5,000 times. A difference is called significant when its 95% interval excludes zero. Ours (512) is tested with its first seed; its three seeds differ by at most 0.001 in mean S_α under corruption on COD10K.

II. Held-out corruptions per corruption

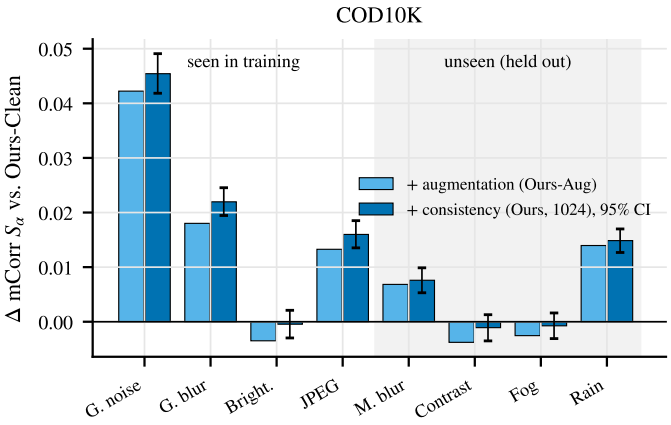


Fig. S1. Gain of corruption augmentation (Ours-Aug) and of the full model (Ours, 1024) over Ours-Clean, per corruption on COD10K. Error bars: 95% bootstrap CI of the full model's gain. Unseen corruptions are shaded.

III. Robustness on the other datasets

IV. Per-corruption Dice retention

V. Accuracy and retention against latency

VI. Further qualitative results

TABLE S1

PAIRED BOOTSTRAP TESTS (5,000 RESAMPLES): MEAN PER-IMAGE DIFFERENCE IN S_α UNDER CORRUPTION AND ITS 95% CI. BOLD: CI EXCLUDES ZERO. GROUPS: ALL 40 CORRUPTED CONDITIONS UNLESS STATED; SEEN/UNSEEN REFER TO THE CORRUPTIONS USED / NEVER USED IN OUR TRAINING.

Comparison	COD10K (n=2026)	NC4K (n=4121)	CAMO (n=250)	CHAMELEON (n=76)
Ours (1024) − SINet-V2	**+0.017 [+0.013, +0.021]**	**+0.009 [+0.007, +0.012]**	-0.006 [-0.019, +0.007]	**-0.024 [-0.038, -0.010]**
Ours (1024) − ZoomNet	**+0.009 [+0.005, +0.013]**	**+0.013 [+0.011, +0.016]**	**+0.014 [+0.001, +0.028]**	**-0.020 [-0.033, -0.007]**
Ours (1024) − PFNet	**+0.039 [+0.035, +0.043]**	**+0.034 [+0.032, +0.037]**	**+0.039 [+0.025, +0.053]**	-0.002 [-0.019, +0.016]
Ours (1024) − SAM2-UNet	**-0.054 [-0.058, -0.050]**	**-0.052 [-0.054, -0.049]**	**-0.075 [-0.088, -0.063]**	**-0.049 [-0.066, -0.033]**
Ours (1024) − SAM-Adapter	**-0.041 [-0.045, -0.037]**	–	**-0.026 [-0.038, -0.014]**	-0.010 [-0.025, +0.006]
Ours (512) − SINet-V2	**+0.010 [+0.006, +0.013]**	**+0.018 [+0.016, +0.021]**	+0.008 [-0.005, +0.021]	**-0.038 [-0.058, -0.020]**
Ours (512) − ZoomNet	+0.002 [-0.002, +0.007]	**+0.022 [+0.019, +0.025]**	**+0.028 [+0.014, +0.043]**	**-0.034 [-0.054, -0.016]**
Ours (512) − Ours (1024) (local)	-0.000 [-0.003, +0.002]	**+0.013 [+0.012, +0.015]**	**+0.017 [+0.009, +0.027]**	-0.003 [-0.018, +0.011]
Ours (512) − Ours (1024) (global)	**-0.018 [-0.022, -0.014]**	+0.001 [-0.001, +0.003]	+0.008 [-0.003, +0.019]	**-0.033 [-0.052, -0.016]**
Ours (1024) − Ours-Clean (seen)	**+0.021 [+0.019, +0.023]**	–	**+0.026 [+0.021, +0.031]**	**+0.024 [+0.014, +0.034]**
Ours (1024) − Ours-Clean (unseen)	**+0.005 [+0.003, +0.007]**	–	**+0.010 [+0.005, +0.015]**	+0.007 [-0.003, +0.016]
Ours (1024) − Ours-Clean (rain)	**+0.015 [+0.013, +0.017]**	–	**+0.014 [+0.008, +0.021]**	+0.012 [-0.000, +0.025]

TABLE S2

ROBUSTNESS ON NC4K (8 CORRUPTIONS × 5 SEVERITIES). CLEAN AND MCORR: S_α; RET.: MCORR/CLEAN (S_α UNLESS MARKED); SEEN/UNSEEN: RETENTION ON THE CORRUPTIONS USED / NEVER USED TO TRAIN OUR MODELS; LOCAL: NOISE, BLUR, JPEG, RAIN; GLOBAL: BRIGHTNESS, CONTRAST, FOG; MCE: CORRUPTION ERROR RELATIVE TO SINET-V2. INPUT: SIDE LENGTH IN PX (ZOOMNET* ALSO USES 0.5× AND 1.5×). ‡ORACLE WITH THE GROUND-TRUTH BOX, NOT A COMPETITOR. RETENTION ALONE CAN FAVOUR WEAKER MODELS; READ IT WITH MCORR AND MCE.

Model	Input	Clean	mCorr	Ret.	$\text{Ret.}_{\text{Dice}}$	Seen	Unseen	Local	Global	mCE↓
SINet-V2	352	0.847	0.826	0.975	0.963	0.980	0.971	0.961	**0.999**	1.00
PFNet	416	0.829	0.801	0.967	0.947	0.969	0.964	0.956	0.985	1.15
ZoomNet	384*	0.853	0.822	0.964	0.940	0.965	0.962	0.951	0.985	1.02
SAM2-UNet	352	**0.901**	**0.887**	**0.985**	**0.978**	**0.984**	**0.986**	**0.978**	0.996	**0.65**
SAM-Adapter	1024	–	–	–	–	–	–	–	–	–
Ours (1024)	1024	0.859	0.836	0.973	0.961	0.974	0.973	0.958	**0.999**	0.95
Ours (512)	512	0.861	0.844	0.980	0.972	0.980	0.980	0.970	0.996	0.90
MobileSAM + GT box‡	1024	0.850	0.831	0.978	0.974	0.976	0.979	0.968	0.995	0.97

TABLE S3

ROBUSTNESS ON CAMO (8 CORRUPTIONS × 5 SEVERITIES). CLEAN AND MCORR: S_α; RET.: MCORR/CLEAN (S_α UNLESS MARKED); SEEN/UNSEEN: RETENTION ON THE CORRUPTIONS USED / NEVER USED TO TRAIN OUR MODELS; LOCAL: NOISE, BLUR, JPEG, RAIN; GLOBAL: BRIGHTNESS, CONTRAST, FOG; MCE: CORRUPTION ERROR RELATIVE TO SINET-V2. INPUT: SIDE LENGTH IN PX (ZOOMNET* ALSO USES 0.5× AND 1.5×). ‡ORACLE WITH THE GROUND-TRUTH BOX, NOT A COMPETITOR. RETENTION ALONE CAN FAVOUR WEAKER MODELS; READ IT WITH MCORR AND MCE.

Model	Input	Clean	mCorr	Ret.	$\text{Ret.}_{\text{Dice}}$	Seen	Unseen	Local	Global	mCE↓
SINet-V2	352	0.820	0.791	0.964	0.948	0.966	0.962	0.946	**0.996**	1.00
PFNet	416	0.782	0.746	0.953	0.921	0.951	0.955	0.940	0.974	1.22
ZoomNet	384*	0.820	0.771	0.940	0.897	0.941	0.939	0.919	0.974	1.10
SAM2-UNet	352	**0.884**	**0.860**	**0.973**	**0.961**	**0.969**	**0.978**	**0.963**	0.990	**0.67**
SAM-Adapter	1024	0.847	0.811	0.958	0.928	0.947	0.969	0.937	0.994	0.90
Ours (1024)	1024	0.816	0.785	0.962	0.945	0.958	0.967	0.943	0.994	1.03
Ours (512)	512	0.825	0.799	0.968	0.951	0.966	0.969	0.954	0.992	0.97
MobileSAM + GT box‡	1024	0.827	0.804	0.972	0.968	0.970	0.974	0.960	0.991	0.95

TABLE S4

ROBUSTNESS ON CHAMELEON (8 CORRUPTIONS × 5 SEVERITIES). CLEAN AND MCORR: S_α; RET.: MCORR/CLEAN (S_α UNLESS MARKED); SEEN/UNSEEN: RETENTION ON THE CORRUPTIONS USED / NEVER USED TO TRAIN OUR MODELS; LOCAL: NOISE, BLUR, JPEG, RAIN; GLOBAL: BRIGHTNESS, CONTRAST, FOG; MCE: CORRUPTION ERROR RELATIVE TO SINET-V2. INPUT: SIDE LENGTH IN PX (ZOOMNET* ALSO USES 0.5× AND 1.5×). ‡ORACLE WITH THE GROUND-TRUTH BOX, NOT A COMPETITOR. RETENTION ALONE CAN FAVOUR WEAKER MODELS; READ IT WITH MCORR AND MCE.

Model	Input	Clean	mCorr	Ret.	$\text{Ret.}_{\text{Dice}}$	Seen	Unseen	Local	Global	mCE↓
SINet-V2	352	0.888	0.870	**0.980**	**0.971**	**0.986**	0.975	0.968	**1.000**	1.00
PFNet	416	0.882	0.849	0.963	0.942	0.967	0.958	0.950	0.983	1.17
ZoomNet	384*	0.902	0.867	0.961	0.938	0.962	0.960	0.944	0.989	1.03
SAM2-UNet	352	**0.914**	**0.896**	**0.980**	**0.971**	0.978	**0.982**	**0.972**	0.994	**0.81**
SAM-Adapter	1024	0.896	0.857	0.956	0.922	0.947	0.965	0.937	0.989	1.11
Ours (1024)	1024	0.883	0.847	0.959	0.940	0.957	0.962	0.938	0.995	1.18
Ours (512)	512	0.851	0.833	0.978	0.969	0.978	0.979	0.970	0.992	1.31
MobileSAM + GT box‡	1024	0.787	0.762	0.968	0.962	0.966	0.971	0.954	0.993	1.86

TABLE S5

DICE RETENTION PER CORRUPTION ON COD10K (MEAN OVER SEVERITIES 1–5 DIVIDED BY CLEAN DICE). ^{s}SEEN / ^{u}UNSEEN IN OUR TRAINING. RAIN USES OUR OWN SYNTHETIC STREAKS. OURS (512): MEAN ± STD OVER 3 SEEDS.

Model	G. noise^s	M. blur^u	G. blur^s	Bright.^{s}	Contrast^u	Fog^u	JPEG^s	Rain^u
SINet-V2	0.938	0.927	0.938	0.996	**0.993**	0.994	0.983	0.881
PFNet	0.932	0.917	0.908	0.963	0.958	0.965	**0.988**	0.866
ZoomNet	0.883	0.900	0.873	0.960	0.958	0.969	0.970	0.828
SAM2-UNet	**0.963**	**0.971**	**0.961**	0.988	0.986	**0.995**	0.981	**0.940**
SAM-Adapter	0.876	0.928	0.892	0.977	0.979	0.993	0.944	0.908
Ours (1024)	0.904	0.935	0.929	**0.997**	0.989	0.992	0.952	0.869
Ours (512)	$0.939_{\pm.001}$	$0.956_{\pm.002}$	$0.946_{\pm.002}$	$0.990_{\pm.000}$	$0.985_{\pm.001}$	$0.990_{\pm.002}$	$0.978_{\pm.001}$	$0.914_{\pm.003}$

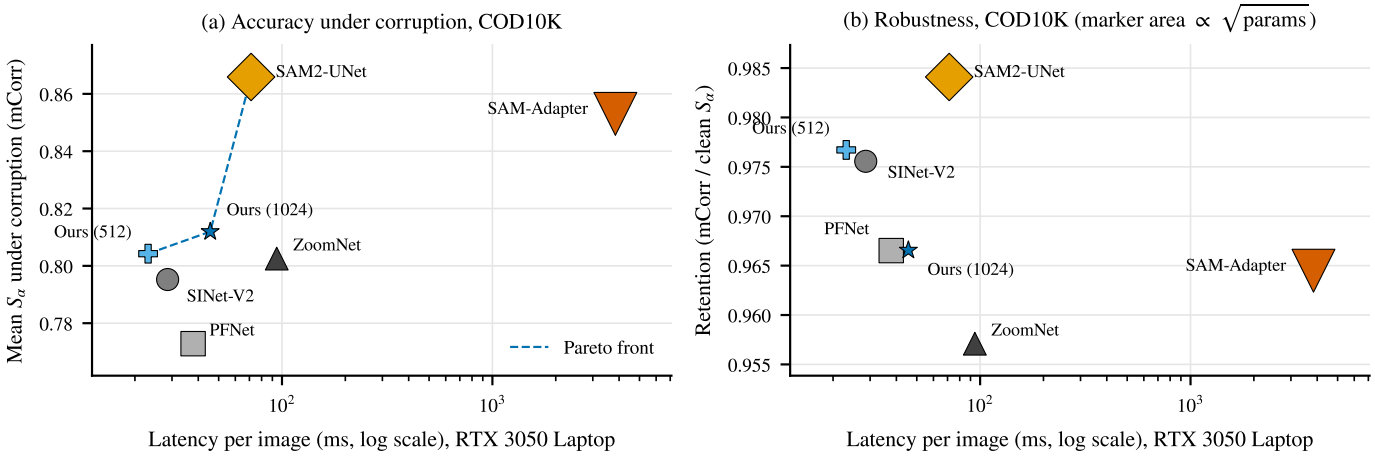


Fig. S2. Accuracy under corruption (a) and retention (b) against latency on COD10K. Retention alone ranks some weaker models above stronger ones; read it together with (a).

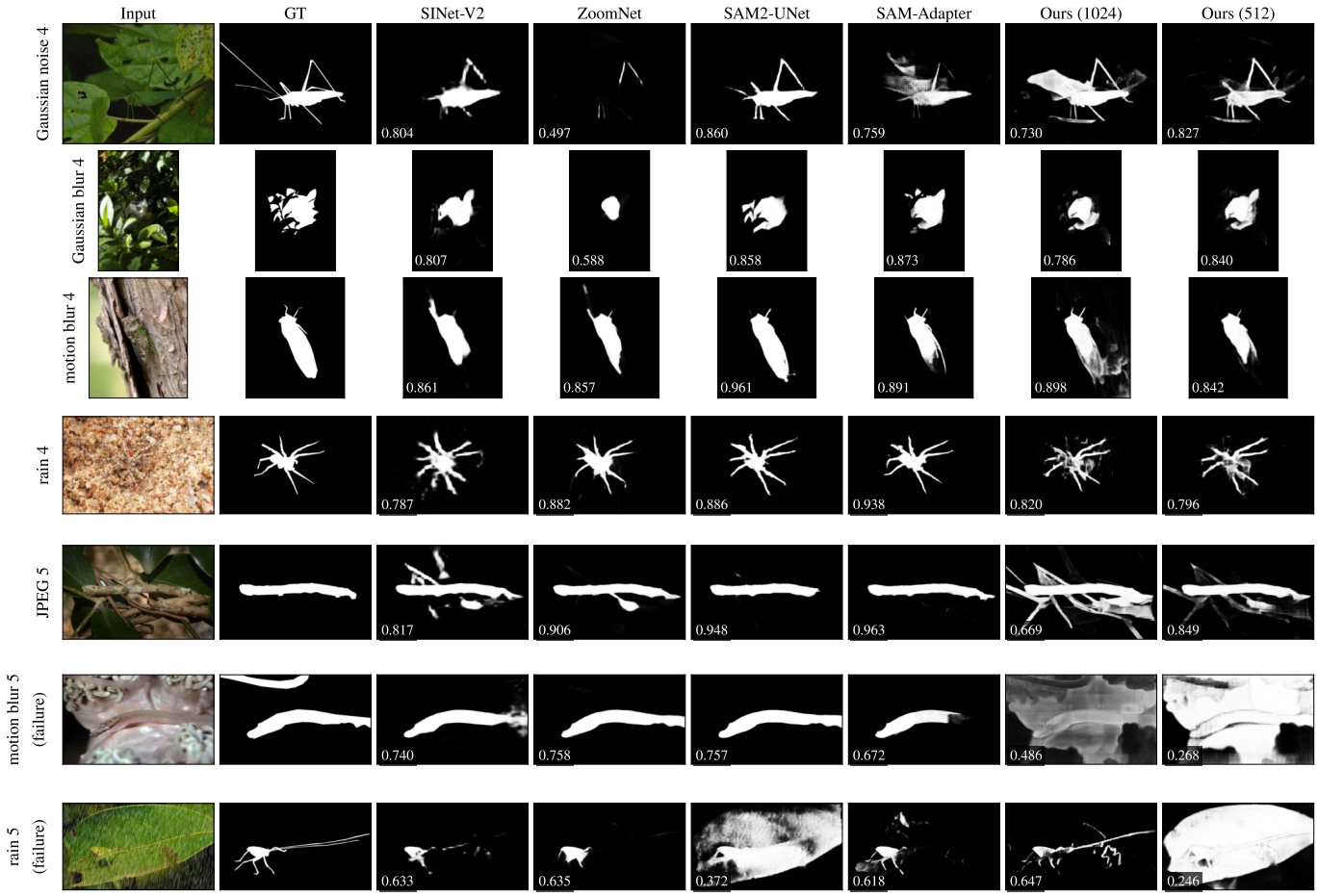


Fig. S3. All rows of the qualitative selection. For each corruption: the image on which Ours (512) has its median S_α; the last two rows: its lowest-scoring images. Images and scores are listed in results/qualitative_selection.csv.